# PacoMouseCYD by Paco

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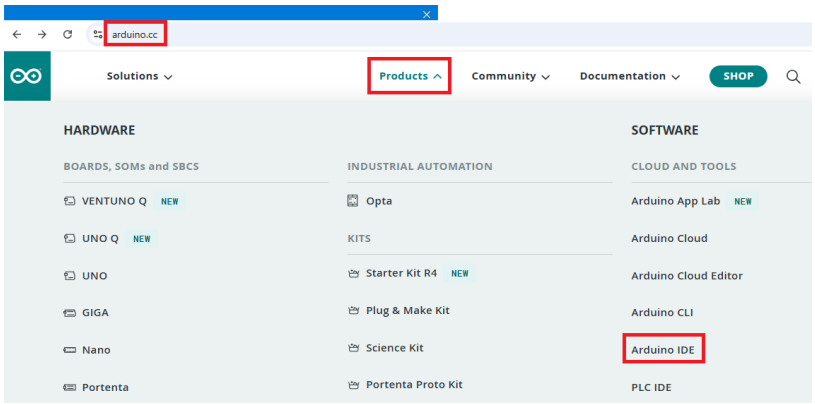
|---|---|
| EN | *PacoMouseCYD for beginners* |
| ES | *PacoMouseCYD para novatos* |
| DE | *PacoMouseCYD für Anfänger* |
| CZ | *PacoMouseCYD pro začátečníky* |

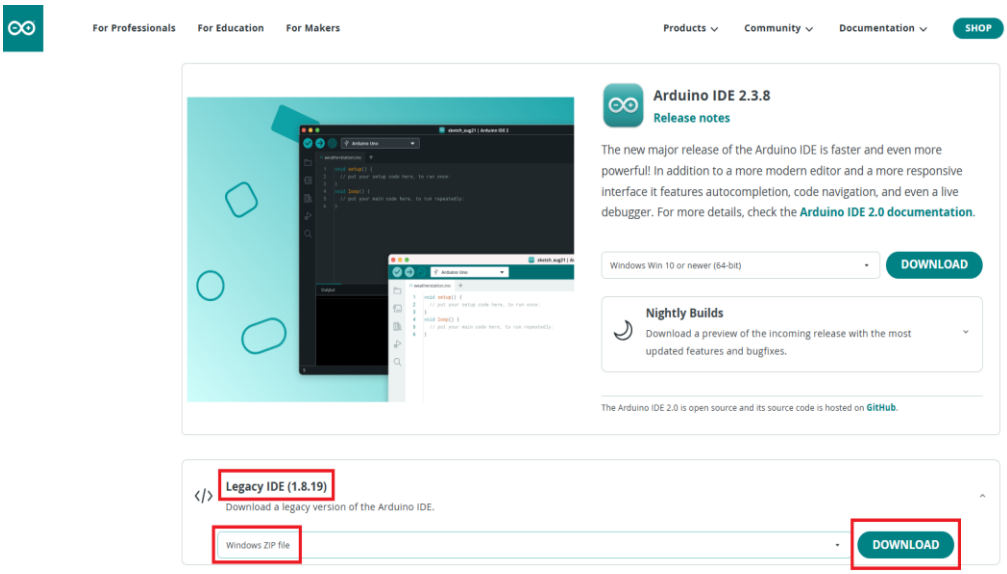
v0.11 04/2026

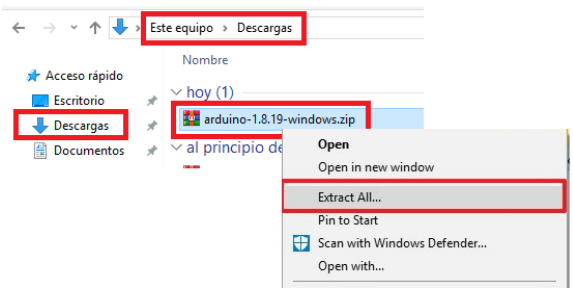
Paco Cañada https://usuaris.tinet.cat/fmco/

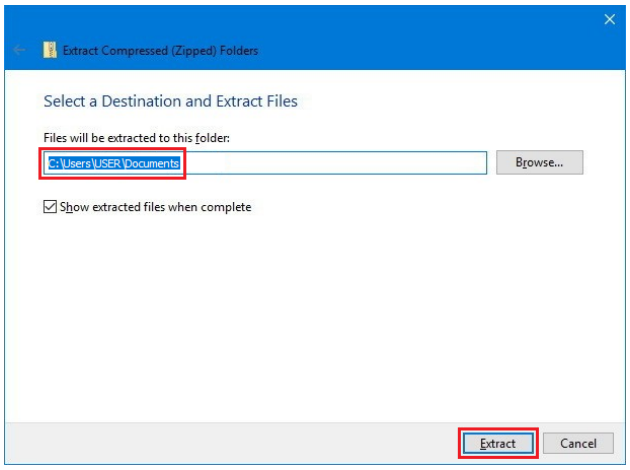
| | |
|---|---|
| EN | **Step 1 – Download Arduino IDE** |
| ES | **Paso 1 – Descargar Arduino IDE** |
| DE | **Schritt 1 – Arduino IDE herunterladen** |
| CZ | **Krok 1 – Stáhněte si Arduino IDE** |

https://www.arduino.cc/



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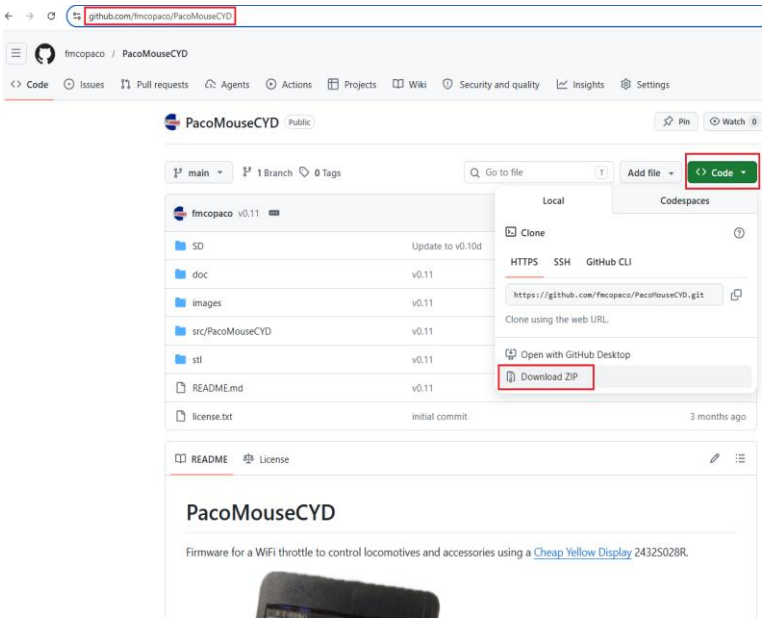
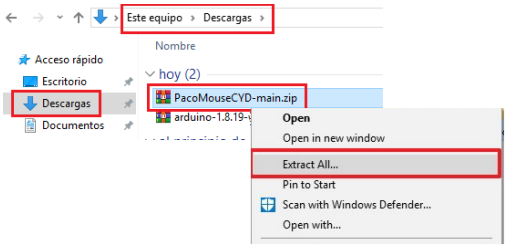
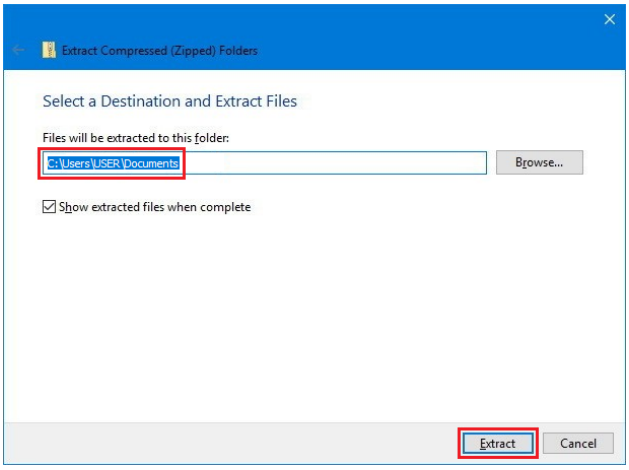
|---|---|
| EN | **Step 2 – Download PacoMouseCYD** |
| ES | **Paso 2 – Descargar PacoMouseCYD** |
| DE | **Schritt 2 – PacoMouseCYD herunterladen** |
| CZ | **Krok 2 – Stáhněte si PacoMouseCYD** |

https://github.com/fmcopaco/PacoMouseCYD



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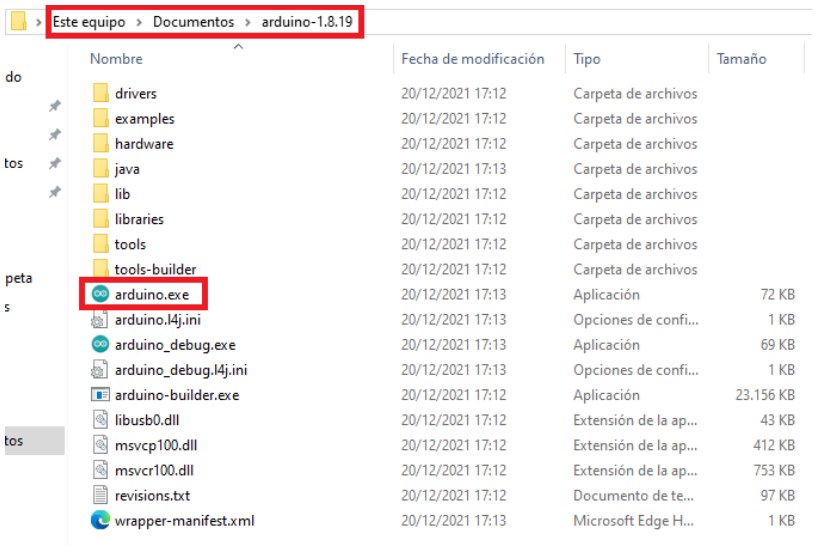
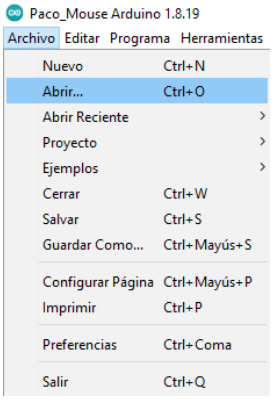
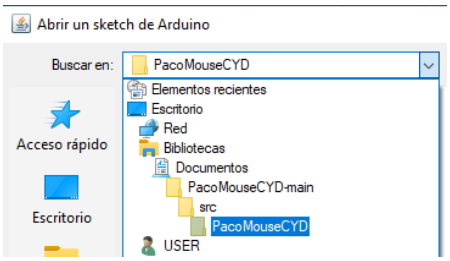
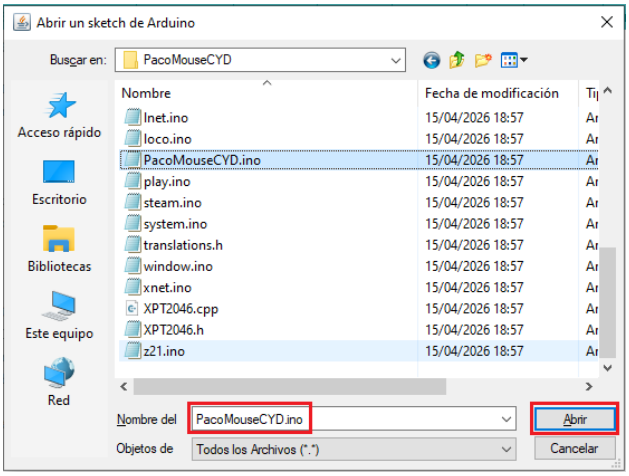
|---|---|
| EN | **Step 3 – Open PacoMouseCYD.ino in Arduino IDE** |
| ES | **Paso 3 – Abrir PacoMouseCYD.ino en Arduino IDE** |
| DE | **Schritt 3 – Öffnen Sie PacoMouseCYD.ino in der Arduino IDE** |
| CZ | **Krok 3 – Otevřete PacoMouseCYD.ino v Arduino IDE** |



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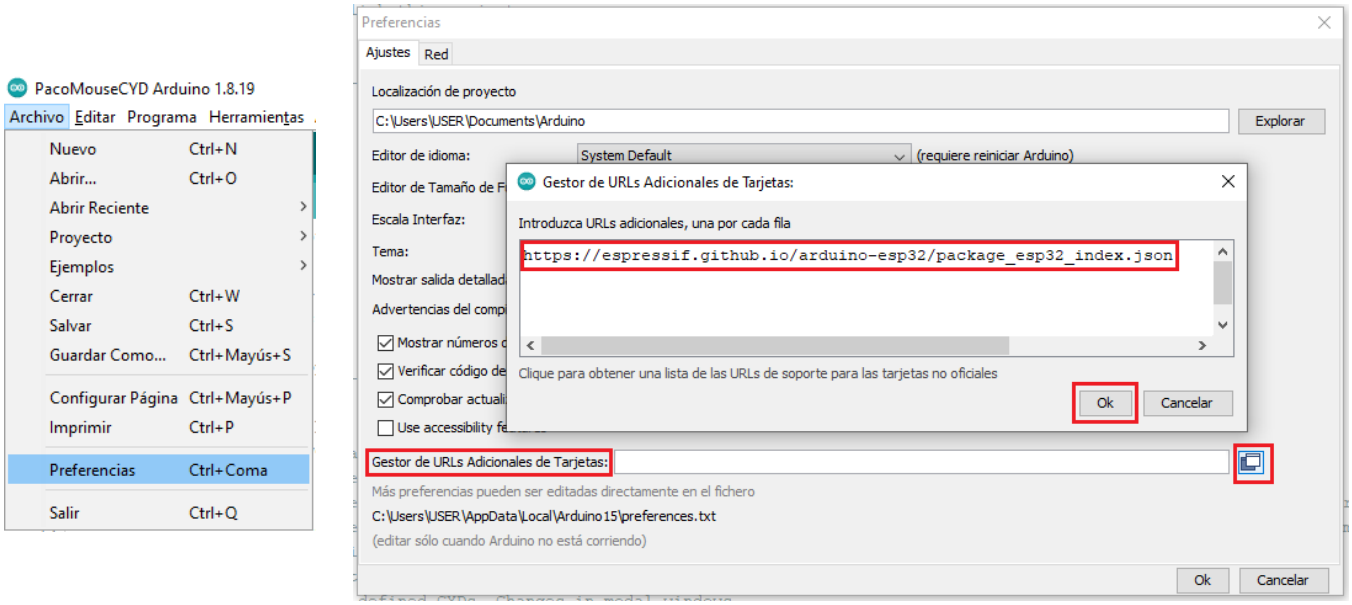
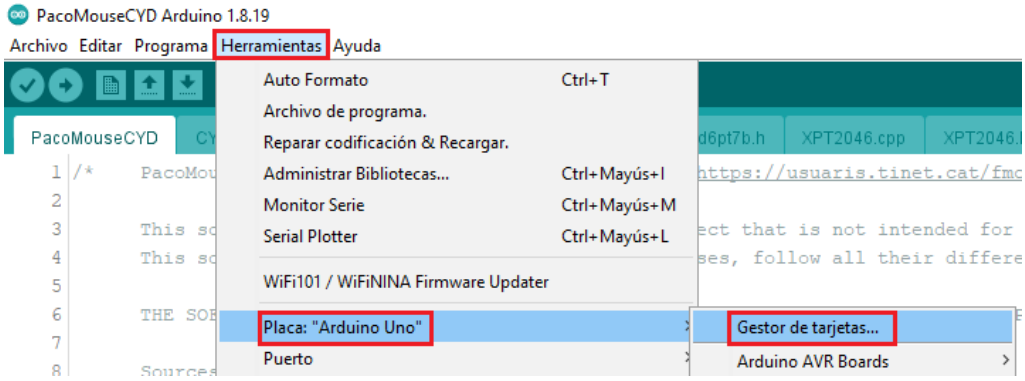
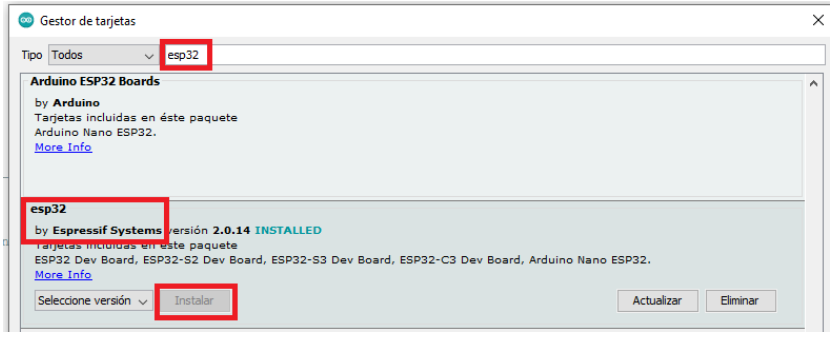
|---|---|
| EN | **Step 4 – Install Arduino core for ESP32** |
| ES | **Paso 4 – Instalar Arduino core para ESP32** |
| DE | **Schritt 4 – Arduino-Core für ESP32 installieren** |
| CZ | **Krok 4 – Instalace jádra Arduina pro ESP32** |

**https://espressif.github.io/arduino-esp32/package_esp32_index.json**

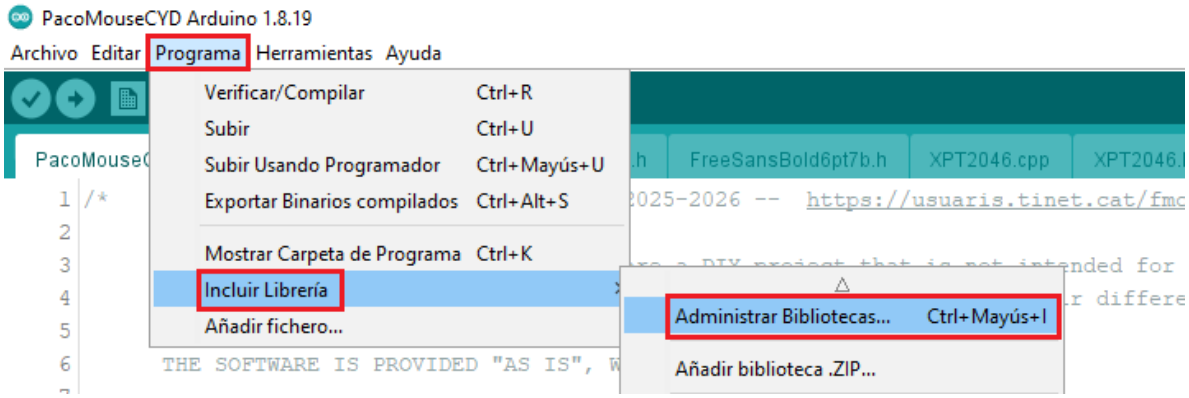


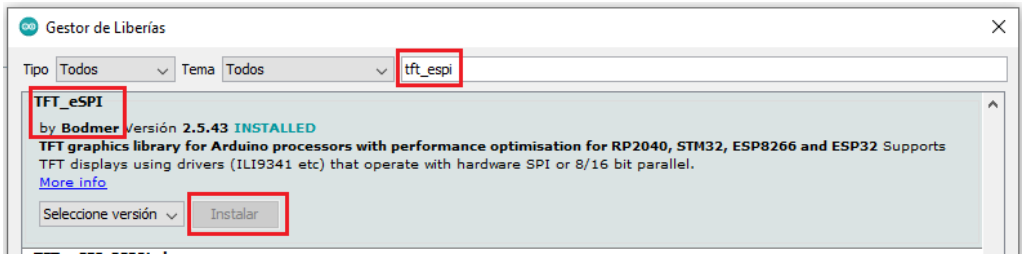

***ESP32 by Espressif Systems (v2.0.14 o v3.3.0)***



| | |
|---|---|
| EN | **Step 5 – Install TFT_eSPI library** |
| ES | **Paso 5 – Instalar librería TFT_eSPI** |
| DE | **Schritt 5 – Installieren Sie die TFT_eSPI-Bibliothek** |
| CZ | **Krok 5 – Instalace knihovny TFT_eSPI** |

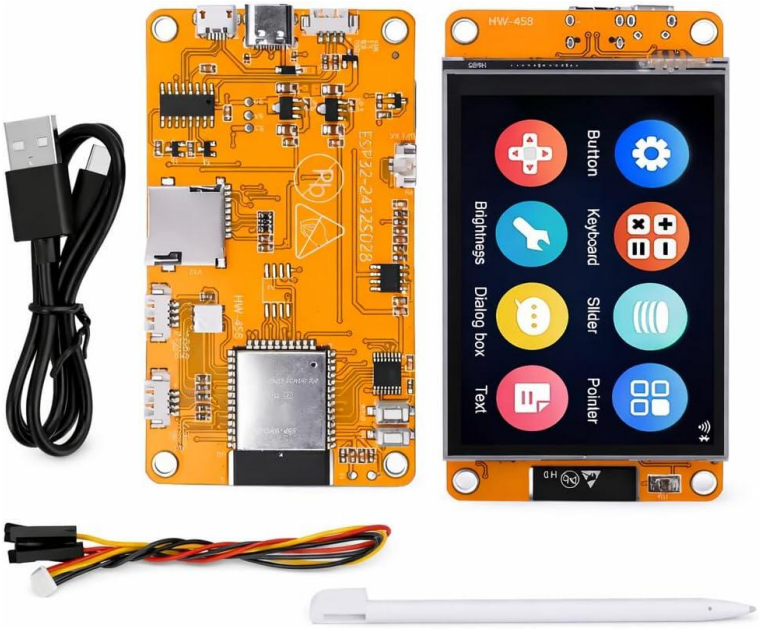


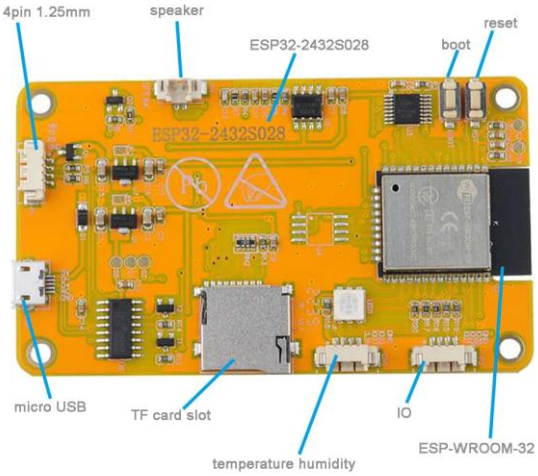

***TFT_eSPI.h by Bodmer(v2.5.43)***



| | |
|---|---|
| EN | **Step 6 – Update the User_Setup.h file for your Cheap Yellow Display** |
| ES | **Paso 6 – Actualizar el archivo User_Setup.h para su Cheap Yellow Display** |
| DE | **Schritt 6 – Aktualisieren Sie die Datei User_Setup.h für Ihr Cheap Yellow Display** |
| CZ | **Krok 6 – Aktualizujte soubor User_Setup.h pro váš levný žlutý displej** |

https://github.com/witnessmenow/ESP32-Cheap-Yellow-Display/tree/main

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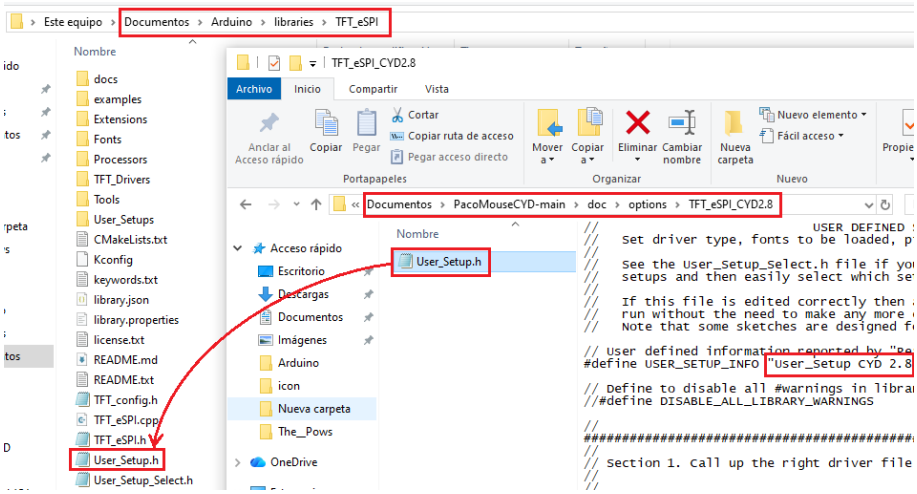
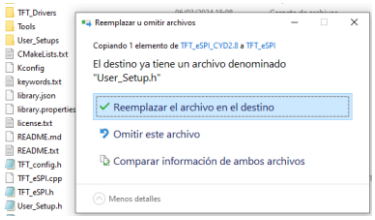
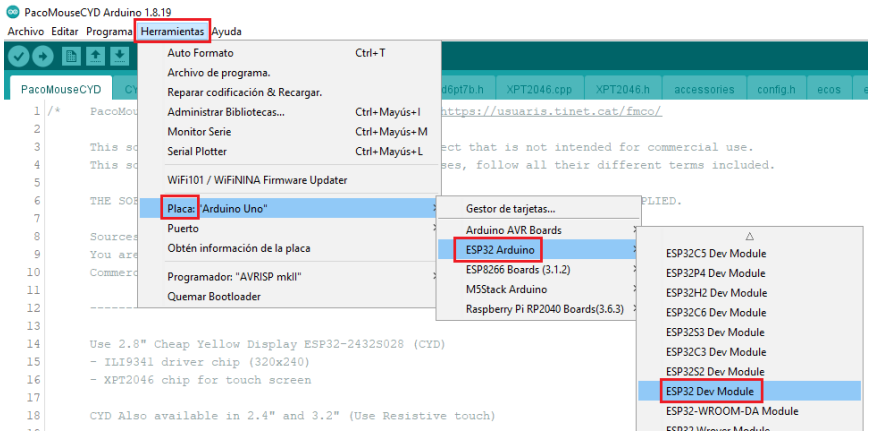
|---|---|---|---|
| ***Cheap Yellow Display*** | Model: **2432S028R** | | |
| **ESP32** | **TFT 2,8”** (320x240 pixel), **ILI9341** | **XPT2406** (**resistive** touch screen) | **SD** ≤ 32Gb (FAT32) |



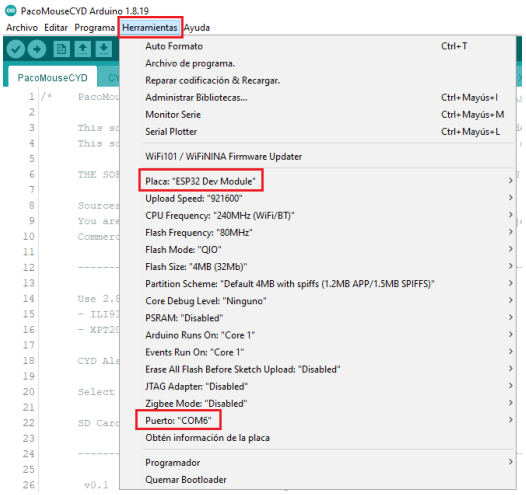



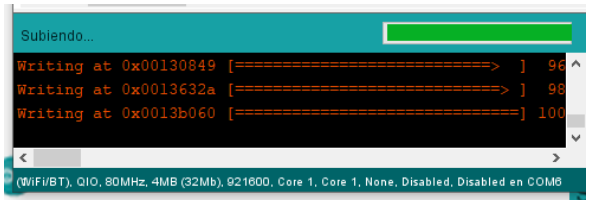
| | |
|---|---|
| EN | **Step 7 – Programming the PacoMouseCYD firmware** |
| ES | **Paso 7 – Programación del firmware del PacoMouseCYD** |
| DE | **Schritt 7 – Programmieren der PacoMouseCYD-Firmware** |
| CZ | **Krok 7 – Programování firmwaru PacoMouseCYD** |

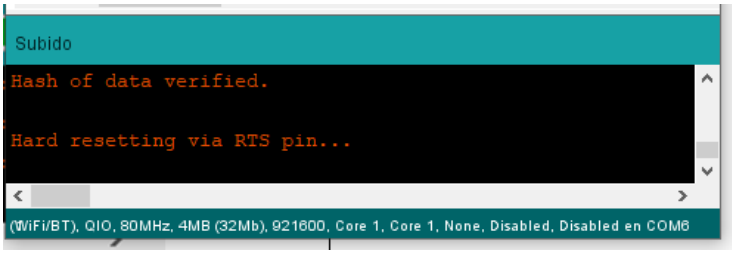
***ESP32 Dev Module***

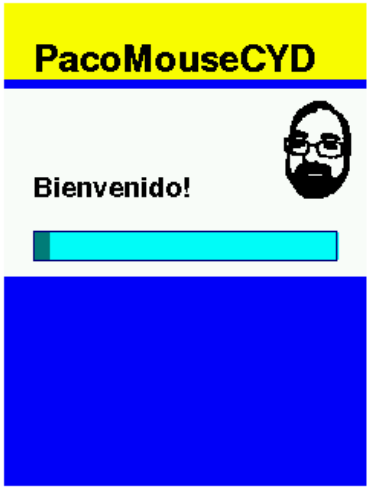



**Driver USB CH340** ([https://www.wch-ic.com/downloads/CH341SER_ZIP.html](https://www.wch-ic.com/downloads/CH341SER_ZIP.html))

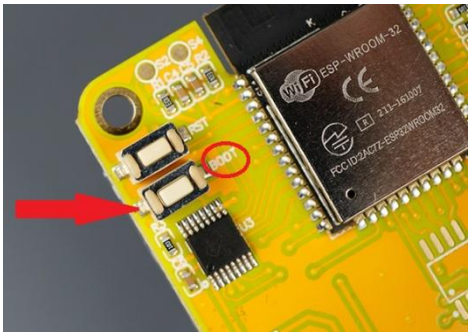


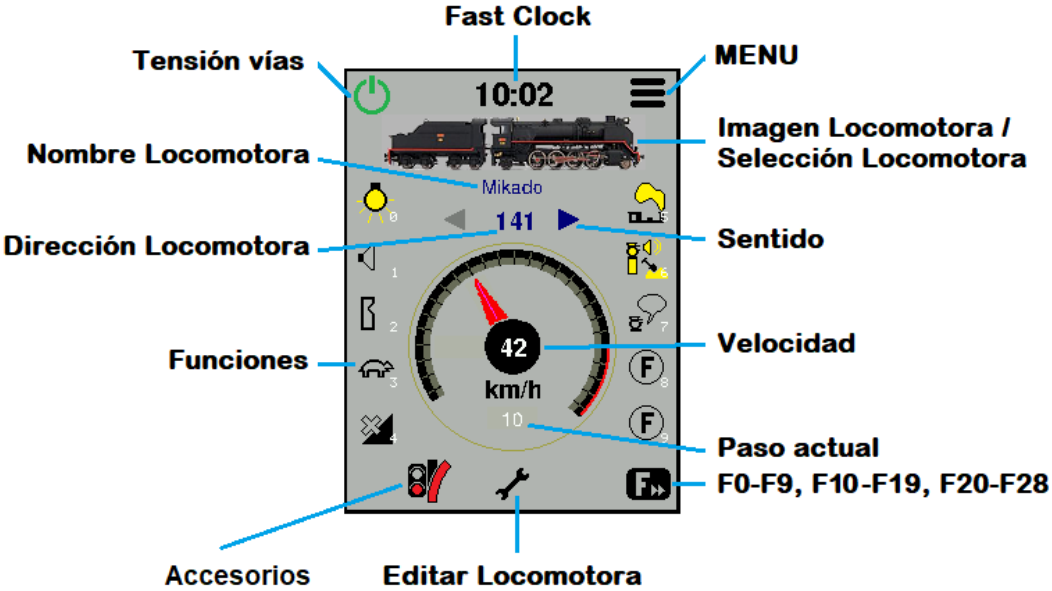


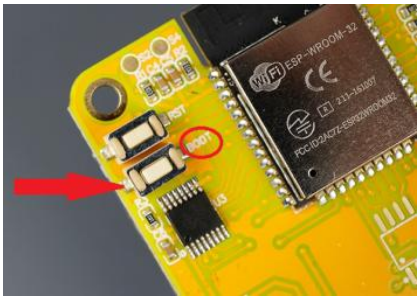

| | |
|---|---|
| EN | If you receive an error message stating that the chip is not in *Download mode*, press the BOOT button until the program starts uploading. |
| ES | Si aparece el error de que el chip no está en *Download mode* pulsar el botón BOOT hasta que empiece a subir el programa |
| DE | Wenn Sie eine Fehlermeldung erhalten, die besagt, dass sich der Chip nicht im *Download-Modus* befindet, drücken Sie die BOOT-Taste so lange, bis das Programm mit dem Hochladen beginnt. |
| CZ | Pokud se zobrazí chybová zpráva, že čip není v režimu stahování (*Download*), stiskněte tlačítko BOOT, dokud se program nezačne nahrávat. |



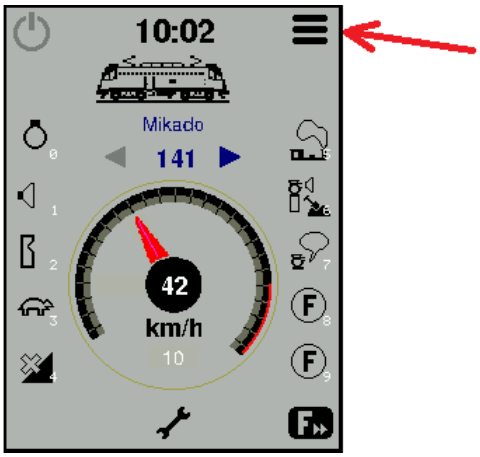



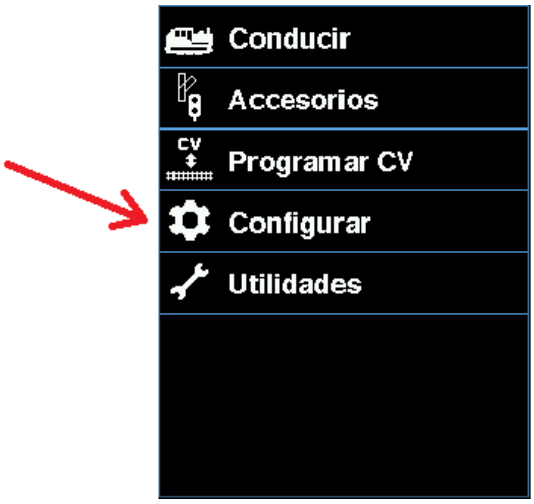
| | |
|---|---|
| EN | **Step 8 – Calibrate the touch screen** |
| ES | **Paso 8 – Calibrar la pantalla táctil** |
| DE | **Schritt 8 – Touchscreen kalibrieren** |
| CZ | **Krok 8 – Kalibrace dotykové obrazovky** |

| | |
|---|---|
| EN | Press BOOT button.<br>Touch to start...<br>To calibrate the touchscreen, touch the corners that appear with the pen. |
| ES | Pulsa el botón BOOT.<br>Toca para iniciar...<br>Para calibrar la pantalla táctil, toca las esquinas que aparecen con el lápiz. |
| DE | Drücken Sie die BOOT-Taste.<br>Tippen Sie zum Starten...<br>Um den Touchscreen zu kalibrieren, berühren Sie die erscheinenden Ecken mit dem Stift. |
| CZ | Stiskněte tlačítko BOOT.<br>Dotykem spusťte...<br>Chcete-li kalibrovat dotykovou obrazovku, dotkněte se perem zobrazených rohů. |



| | |
|---|---|
| EN | **Step 9 – Select language** |
| ES | **Paso 9 – Seleccionar idioma** |
| DE | **Schritt 9 – Sprache auswählen** |
| CZ | **Krok 9 – Vyberte jazyk** |



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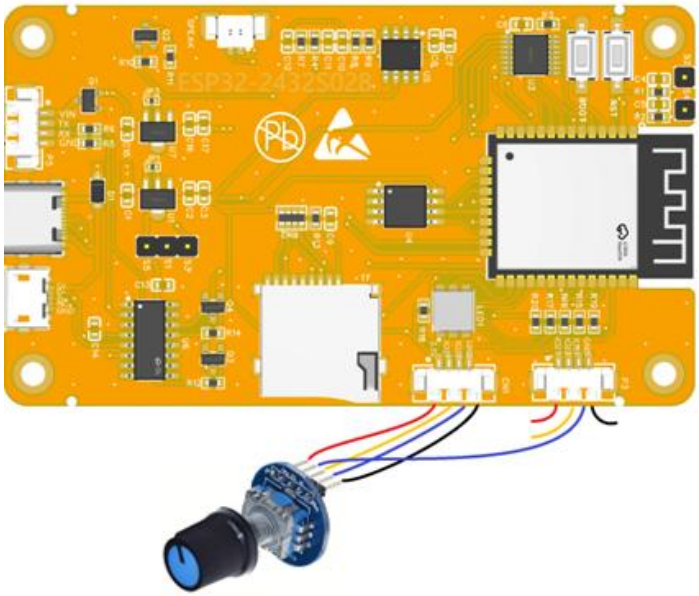
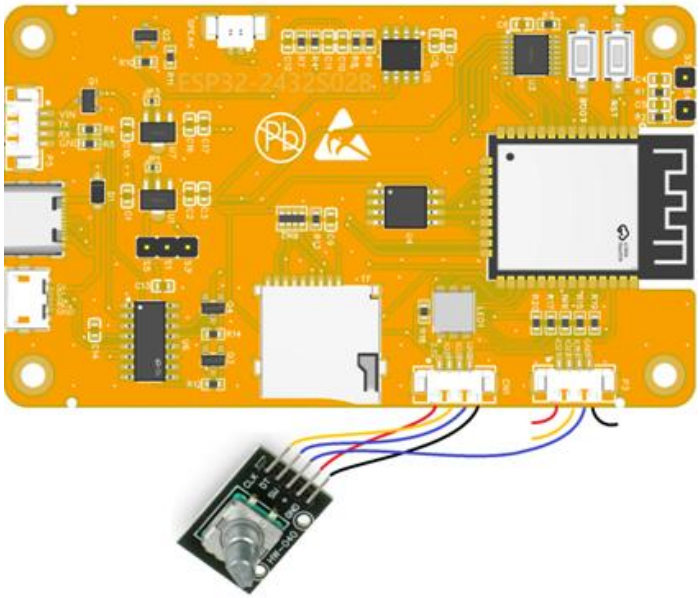
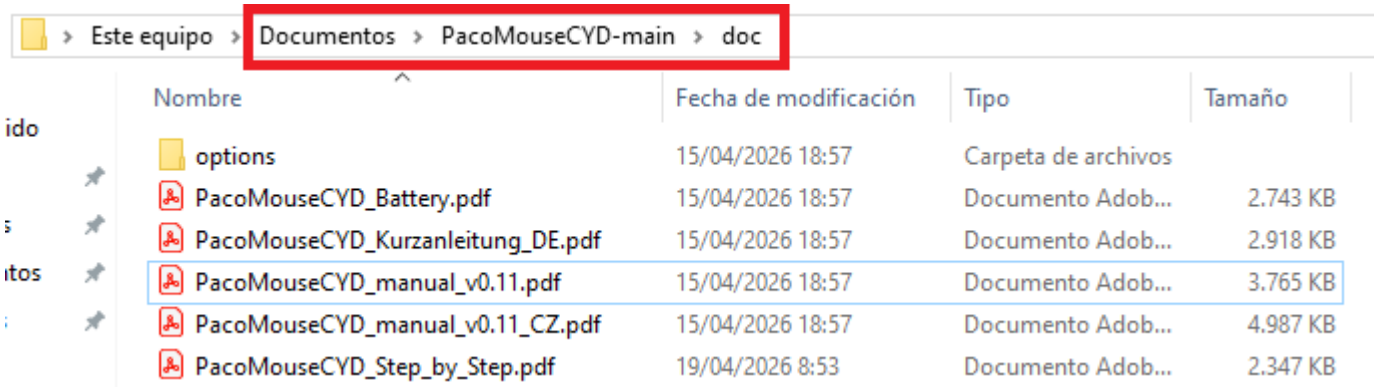
|---|---|
| EN | **Step 10 – Connect encoder** |
| ES | **Paso 10 – Conectar encoder** |
| DE | **Schritt 10 – Encoder anschließen** |
| CZ | **Krok 10 – Připojení enkodéru** |

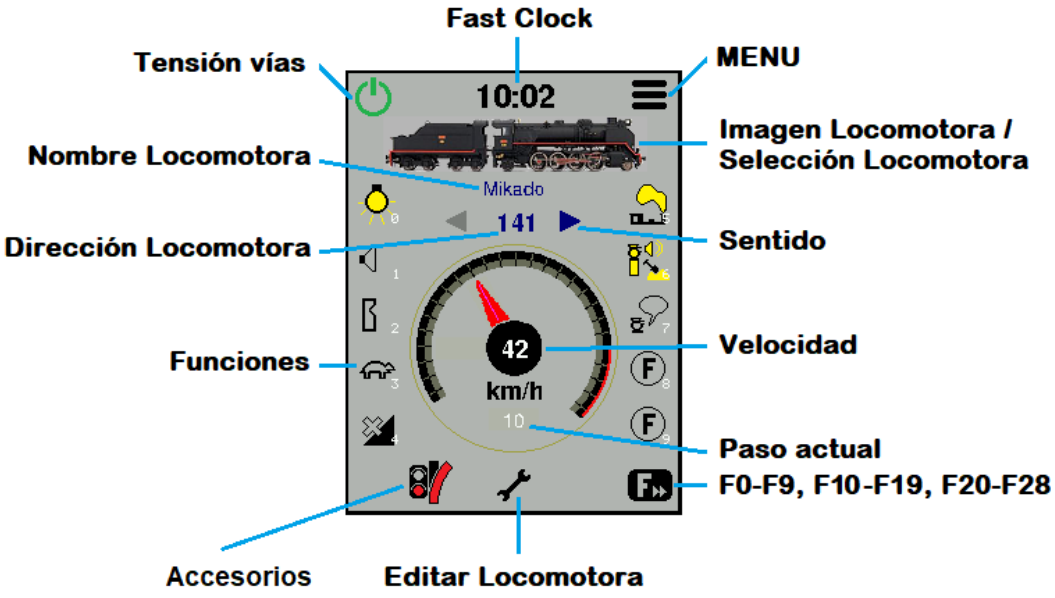


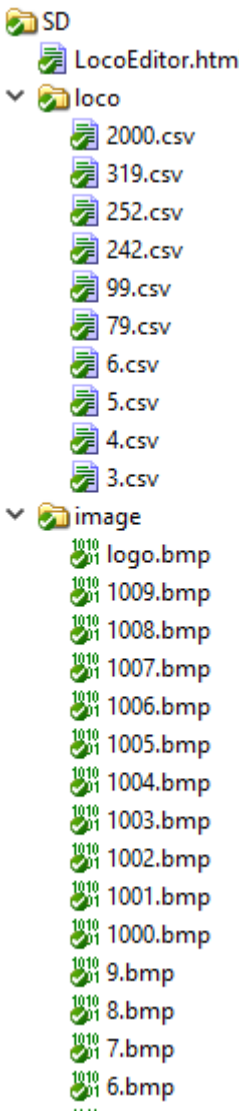
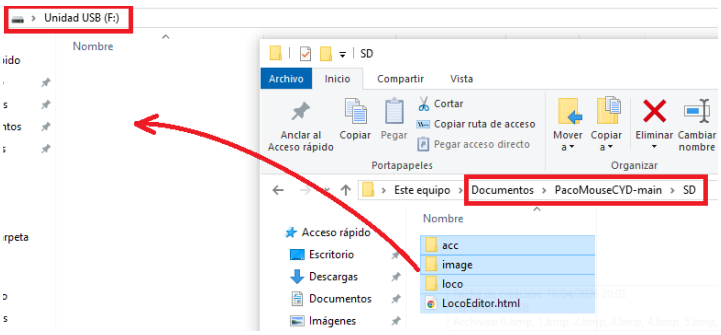


| ENCODER | | CYD | Conector |
|---|---|---|---|
| GND | GND | GND | CN1 |
| S1 | CLK | IO22 | CN1 |
| S2 | DT | IO27 | CN1 |
| Key | SW | IO35 | P3 |
| 5V | + | +3V3 | CN1 |

| | |
|---|---|
| EN | **READ THE MANUAL !!!** |
| ES | **¡ LEA EL MANUAL !** |
| DE | **LESEN SIE DIE ANLEITUNG!!!** |
| CZ | **PŘEČTĚTE SI NÁVOD !!!** |



| | |
|---|---|
| EN | **ANNEX: SD CARD** |
| ES | **ANEXO: TARJETA SD** |
| DE | **ANHANG: SD-KARTE** |
| CZ | **PŘÍLOHA: SD KARTA** |

| | |
|---|---|
| EN | You can use **PacoMouseCYD** without an SD card; however, the color locomotive images will not be available.<br>Use an SD card with a maximum capacity of 32GB (**FAT32** format). Insert the SD card before turning on **PacoMouseCYD**; otherwise, the internal hard drive of the *Cheap Yellow Display* will be used for the various data.<br>The SD card stores the color images of the locomotives, as well as their data (with the ECoS protocol, only the images are used, as the data is provided by the control unit) and the data for the accessory panels. |
| ES | Se puede usar **PacoMouseCYD** sin una tarjeta SD, en este caso las imágenes de las locomotoras a color no estarán disponibles.<br>Use una tarjeta SD de cómo máximo 32Gb (formato **FAT32**). Inserte la tarjeta SD antes de encender el **PacoMouseCYD** sino se usará el disco interno del *Cheap Yellow Display* en su lugar para los diferentes datos.<br>En la tarjeta se almacenan las imágenes a color de las locomotoras, así como los datos de estas (con protocolo ECoS solo se usan las imágenes ya que los datos los proporciona la central) y de los paneles de accesorios. |
| DE | **PacoMouseCYD** kann auch ohne SD-Karte verwendet werden; die Farbbilder der Lokomotiven sind dann jedoch nicht verfügbar.<br>Verwenden Sie eine SD-Karte mit maximal 32 GB Speicherkapazität (**FAT32**-Format). Legen Sie die SD-Karte ein, bevor Sie **PacoMouseCYD** einschalten; andernfalls werden die Daten auf dem internen Speicher des *Cheap Yellow Displays* gespeichert.<br>Auf der SD-Karte werden die Farbbilder der Lokomotiven sowie deren Daten gespeichert (beim ECoS-Protokoll werden nur die Bilder verwendet, da die Daten von der Steuereinheit bereitgestellt werden) und die Daten für die Zubehörtafeln. |
| CZ | **PacoMouseCYD** můžete používat i bez SD karty; barevné obrázky lokomotiv však nebudou k dispozici.<br>Používejte SD kartu s maximální kapacitou 32 GB (formát **FAT32**). SD kartu vložte před zapnutím **PacoMouseCYD**; jinak bude pro různá data použit interní pevný disk zařízení *Cheap Yellow Display*.<br>SD karta ukládá barevné obrázky lokomotiv a také jejich data (s protokolem ECoS se používají pouze obrázky, protože data poskytuje řídicí jednotka) a data pro panely příslušenství. |